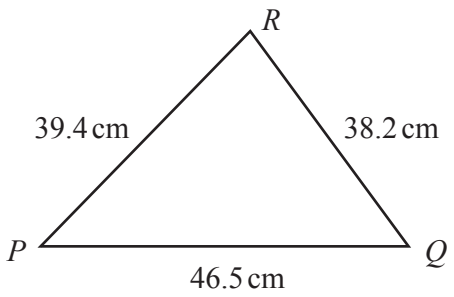
0580_w22_qp_42_Q7

7 **(a)**



NOT TO SCALE

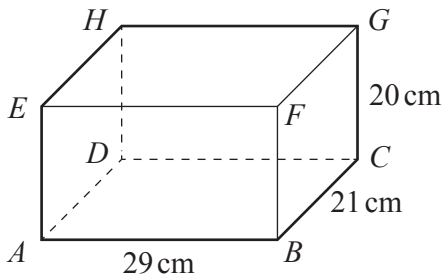
(i) Calculate angle *QPR*.

Angle *QPR* = ... [4]

(ii) Find the shortest distance from *Q* to *PR*.

..... cm [3]

(b) The diagram shows a cuboid.



NOT TO SCALE

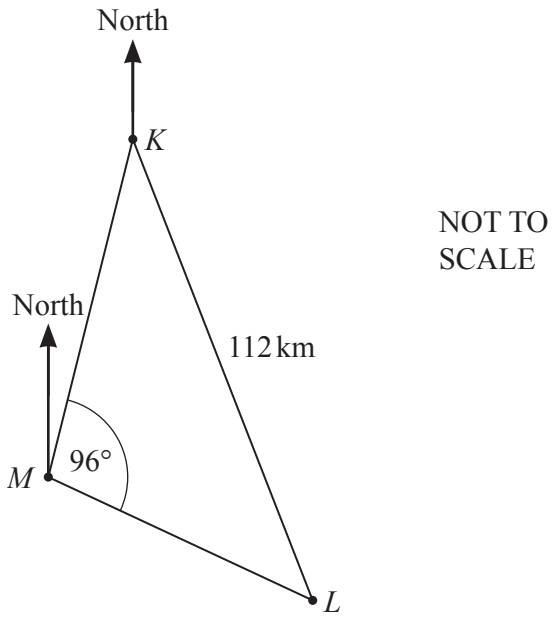
(i) Calculate the length *AG*.

AG = ... cm [3]

(ii) Calculate the angle between AG and the base $ABCD$.

..... [3]

(c)



The diagram shows the positions of a lighthouse, L, and two ships, K and M.
The bearing of L from K is 155° and $KL = 112\,\text{km}$.
The bearing of K from M is 010° and angle $KML = 96°$.

Find the bearing and distance of ship M from the lighthouse, L.

Bearing .....

Distance ..... km [5]